ICT CSA Tutorial 01

01) Briefly explain the modern computer system with its features.

02) What are the two special-purpose registers used for interfacing with primary memory, and what is the primary function of each?

03) What is the main purpose of the Program Counter (PC) in a computer system?

04) What is a "bus" in the context of computer architecture, and what is its fundamental role?

05) Describe the sequence of micro-operations involved in reading data from a specific memory location into a general-purpose register, assuming the address of the data is known.

06) Simulate the execution of the instruction `ADD R1, LOCA` given the following initial conditions and determine the final value of `R1` and `PC`.
Initial Conditions:
The instruction `ADD R1, LOCA` is stored at memory location `1000`.
Initial value of Register `R1` is `50`.
Memory location `LOCA` (which is address `5000`) contains the value `75`.
Initial value of the Program Counter (`PC`) is `1000`.

07) Simulate the execution of the instruction `STORE R2, LOCB` given the following initial conditions and determine the final value stored at `LOCB` and the final value of `PC`.
Initial Conditions:
The instruction `STORE R2, LOCB` is stored at memory location `2000`.
Initial value of Register `R2` is `99`.
Memory location `LOCB` (address `6000`) initially contains `0` (or any arbitrary value, as it will be overwritten).
Initial value of Program Counter (`PC`) is `2000`.
Trace the values of PC, MAR, MDR, IR, and the content of `Mem[LOCB]` through the execution steps.

08) Consider a sequence of two instructions. Trace the PC and the affected register for each step.
Initial Conditions:
Instruction 1: `LOAD R3, 7000` (Load content of memory address `7000` into `R3`) is at memory location `3000`.
Instruction 2: `SUB R3, 10` (Subtract immediate value `10` from `R3`) is at memory location `3004`.
Memory location `7000` contains the value `150`.
Initial value of Register `R3` is `0`.
Initial value of Program Counter (`PC`) is `3000`.

Assume each instruction takes 4 bytes.
Trace the values of PC, MAR, MDR, IR, and R3 through the execution of both instructions.

09) Elaborate on the significance of special-purpose registers (MAR, MDR, PC, IR) in enabling the basic fetch-decode-execute cycle of a computer instruction. How do they collectively ensure the smooth flow of data and control within the processor and memory?

10) Compare and contrast single-bus and multi-bus architectures within a computer processor, highlighting their impact on performance and physical design considerations.